A
Mini Project Report
On

PRIVACY-AWARE PERSONAL DATA STORAGE (P-PDS) LEARNING HOW TO PROTECT USER PRIVACY FROM EXTERNAL APPLICATIONS

Submitted to

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY

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for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
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Certificate

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Certified that this Mini Project Report entitled, **"privacy-aware personal data storage (p-pds) learning how to protect user privacy from external applications"** is the bonafide work of **Meesa Vyshnavi (H.T.No. 22TR1A6635), Rudhroji jyoshna (H.T.No. 22TR1A6652), Samala Shravan Kumar (H.T.No. 22TR1A6653)** and **Kosna Harish Reddy (H.T.No. 22TR1A6627) of III Year, CSE (AI & ML)** in the year 2025 in partial fulfillment of the requirements to award the Degree of Bachelor of Technology in **COMPUTER SCIENCE AND ENGINEERING (AI & ML)** of Sree Chaitanya Institute of Technological Sciences, Karimnagar.

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Finally Special thanks to our parents, sisters and brothers for their support and encouragement throughout my life and this course. Thanks to all our friends and wellwishers for their constant support.

DECLARATION

We hereby declare that the work which is being presented in this report entitled, "**privacy-aware personal data storage (p-pds) learning how to protect user privacy from external applications**", submitted towards the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **CSE (AI&ML)**, Sree Chaitanya Institute of Technological Sciences, Karimnagar is an authentic record of our own work carried out under the supervision of **Mrs.N.Renuka, Assistant Professor, Department of CSE (AI&ML),** Sree Chaitanya Institute of Technological Sciences, Karimnagar.

To the best of our knowledge and belief, this project bears no resemblance with any report submitted to Sree Chaitanya Institute of Technological Sciences or any other University for the award of any degree or diploma.

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ABSTRACT

Recently, Personal Data Storage (PDS) has inaugurated a substantial change to the wag people can store and control their personal data, by moving from a service-centric to a user-centric model. PDS offers individuals the capability to keep their data in a unique logical repository, that can be connected and exploited by proper analytical tools, or shared with third parties under the control of end users. Up to now, most of the research on PDS has focused on how to enforce user privacy preferences and how to secure data when stored into the PDS. In contrast, in this paper we aim at designing a Privacy-aware Personal Data Storage (P-PDS), that is, a PDS able to automatically take privacy-aware decisions on third parties access requests in accordance with user preferences. The proposed P-PDS is based on preliminary results presented in [1], where it has been demonstrated that semi-supervised learning can be successfully exploited to make a PDS able to automatically decide whether an access request has to be authorized or not. In this paper, we have deeply revised the learning process so as to have a more usable P PDS, in terms of reduced effort for the training phase, as well as a more conservative approach

w.r.t. users privacy, when handling conflicting access requests. We run several experiments on a realistic dataset exploiting a group of 360 evaluators. The obtained results show the effectiveness of the proposed approach.

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CHAPTER-1

INTRODUCTION

Nowadays personal data we are digitally producing are scattered in different online systems managed by different providers (e.g., online social media, hospitals, banks, airlines, etc). In this way, on the one hand users are losing control on their data, whose protection is under the responsibility of the data provider, and, on the other, they cannot fully exploit their data, since each provider keeps a separate view of them. To overcome this scenario, Personal Data Storage (PDS) has inaugurated a substantial change to the way people can store and control their personal data, by moving from a service-centric to a user-centric model. PDSs enable individuals to collect into a single logical vault personal information they are producing. Such data can then be connected and exploited by proper analytical tools, as well as shared with third parties under the control of end users. This view is also enabled by recent developments in privacy legislation and, in particular, by the new EU General Data Protection Regulation (GDPR), whose art. 20 states the right to data portability, according to which the data subject shall have the right to receive the personal data concerning him or her, which he or she has provided to a controller, in a structured, commonly used and machine-readable format, thus making possible data collection into a
Up to now, most of the research on PDS has focused on how to enforce user privacy preferences and how to secure data when stored into the PDS . In contrast, the key issue of helping users to specify their privacy preferences on PDS data has not been so far deeply investigated. This is a fundamental issue since average PDS users are not skilled enough to understand how to translate their privacy requirements into a set of privacy preferences. As several studies have shown, average users might have difficulties in properly setting

potentially complex privacy preferences. For example, let us consider Facebooks privacy setting, where users need to configure the options manually according to their desire. authors survey users awareness, attitudes and privacy concerns on profile information and find that only a small number of users change the default privacy preferences on Facebook. Interestingly, in [10], authors find that even when users have changed their default privacy settings, the modified settings do not match the expectations (these are reached only for 39% of users). Moreover, another survey in has shown that Facebook users are not aware enough on protection tools that designed to protect their personal data. According to their study the majority (about 88%) of users had never read the Facebook privacy policy.

To help users on protecting their PDS data, in, we have evaluated the use of different semi-supervised machine learning approaches for learning privacy preferences of PDS owners. The idea is to find a learning algorithm that, after a training period by the PDS owner, returns a classifier able to automatically decide if access requests submitted by third parties are to be authorized or denied. In [1], we have shown that, among different semi-supervised learning approaches, the one that better fits the considered scenario is ensemble learning. Even though the identification of the learning approach is an essential step, the design of a Privacyaware Personal Data Storage (P-PDS), that is, a PDS able to automatically take privacy-aware decisions on third parties access requests requires further investigation. One critical aspect to consider is the usability of the system. Even if semi-supervised techniques require less users effort, compared to manually setting privacy preferences, they still require many interactions with PDS owners to collect a good training dataset.

CHAPTER 2

PROJECT ANALYSIS

2.1 REQUIREMENT SPECIFICATIONS

The software requirements specification specifies the function requirements and non-functional requirements. Functional requirements refer to how the system is going to react according to the input provided and how it is going to behave in particular situations and non-functional requirements refer to Usability, Availability, Reliability, Performance, Security, Supportability, Interface.

H/W System Configuration:-

Processor	-	Ryzen 5000 series
RAM	-	16GB (min)
Hard Disk	-	206B
Key Board	-	Standard Windows Keyboard
Mouse	-	Two or Three Button Mouse
Monitor	-	SVGA

Software Requirements:

Operating System	-	Windows 11
Coding Language	-	Java/J2EE(JSP,Servlet)

Front End	-	J2EE
Back End	-	MySQL

2.2 EXISTING SYSTEM

- Oort [27] is a user-centric cloud storage system that organizes data by users rather than applications, considering global queries which find and combine relevant data fields from relevant users. Moreover, it allows users to choose which applications can access their own data, and which types of data to be shared with which users. Sieve [28] allows userto upload encrypted data to a single cloud storage. It utilizes key-homomorphic scheme to provide cryptographically enforced access control.

- Amber [29] has proposed an architecture where users can choose applications to manipulate their data but it does not mention either how the global queries work or how the application providers interact with. In [2], authors developed a usercentric framework that share with third parity only the answers to a query instead of the raw data. Mortier et al. [30] have proposed a trusted platform called Databox, which can manage personal data by a fine grained access control mechanism but do not focus on policy learning: Recently, [31] proposed a Block chain-based Personal Data Store (BC-PDS) framework, which leverages onBlockChain to secure the storage of personal data. However, all the above proposals focus on access control enforcement, whereas they do not consider user preference or policy learning.

- Privacy preference enforcement have been also investigated in different domains, such as for instance social networks where most of the platforms offer users a privacy setting page to manually set their privacy preferences. Research works have tried to alleviate the burden of this setting' by exploiting machine learning tools. For

instance, [32], [33] have investigated the use of semi-supervised and unsupervised approaches to automatically extract privacy settings in social media. In [34], authors have considered location based data. They have compared the accuracy of manually set privacy preferences with the one of an automated mechanism based on machine learning: The results show that machine learning approaches provide better result than user-defined policies. Bilogrevic et al. [35] also present a privacy preference framework that (semi)automatically predicts sharing decision, based on personal and contextual features. The authors focus only on g location information.

2.2.1Disadvantages

O In the existing work, the system doesn't have strong techniques to implement Privacy-aware Personal Data Storage (P-PDS).

O The system doesn't have active learning which is to select from the training dataset the most representative instances to be labeled by user.

2.3 PROPOSED SYSTEM

The system proposes a revised version of the ensemble learning algorithm proposed in [1], to enforce a more conservative approach w.r.t. users privacy. In particular, we reconsider how ensemble learning handles decisions for access requests for which classifiers return conflicting classes. In general, the final decision is taken selecting the class with the highest aggregated probabilities. However' this presents the limit of not considering user perspective, in that, it does not take into account which classifier is more relevant for the considered user:

To cope with this issue, we propose an alternative strategy for aggregating the class labels returned by the classifiers. According to this approach, we assign a personalized weight to each single classifier used in ensemble learning. We also show how it is possible

to learn these weights from the training dataset, thus without the need of further input from the P-PDS owner: Experiments show that this approach increases users satisfaction as well as the learning effectiveness.

2.3.1Advantages

★ PDS able to automatically take privacy-aware decisions on third parties access requests requires further investigation.

★ The system proposes a revised version of the ensemble learning algorithm proposed in this system, to enforce a more conservative approach w:r.t.users privacy.

2.4 PRELIMINARY INVESTIGATION

The first and foremost strategy for development of a project starts from the thought of designing a mail enabled platform for a small firm in which it is easy and convenient of sending and receiving messages, there is a search engine ,address book and also including some entertaining games. When it is approved by the organization and our project guide the first activity, ie. preliminary investigation begins. The activity has three parts:

Request Clarification

Feasibility Study

Request Approval

REQUEST CLARIFICATION

After the approval of the request to the organization and project guide, with an investigation being considered, the project request must be examined to determine precisely what the system requires.

Here our project is basically meant for users within the company whose systems can be interconnected by the Local Area Network(LAN). In today's busy schedule man need everything should be provided in a readymade manner. So taking into consideration of

the vastly use of the net in day to day life, the corresponding development of the portal came into existence.

2.5 FEASIBILITY ANALYSIS

An important outcome of preliminary investigation is the determination that the system request is feasible. This is possible only if it is feasible within limited resource and time. The different feasibilities that have to be analyzed are

Operational Feasibility

Economic Feasibility

Technical Feasibility

Operational Feasibility

Operational Feasibility deals with the study of prospects of the system to be developed. This system operationally eliminates all the tensions of the Admin and helps him in effectively tracking the project progress. This kind of automation will surely reduce the time and energy, which previously consumed in manual work. Based on the study, the system is proved to be operationally feasible.

Economic Feasibility

Economic Feasibility or Cost-benefit is an assessment of the economic justification for a computer based project. As hardware was installed from the beginning & for lots of purposes thus the cost on project of hardware is low. Since the system is a network based, any number of employees connected to the LAN within that organization can use this tool from at anytime. The Virtual Private Network is to be developed using the existing resources of the organization. So the project is economically feasible.

Technical Feasibility

According to Roger S. Pressman, Technical Feasibility is the assessment of the technical resources of the organization. The organization needs IBM compatible machines with a graphical web browser connected to the Internet and Intranet. The system is developed for platform Independent environment. Java Server Pages, JavaScript, HTML, SQL server and WebLogic Server are used to develop the system. The technical feasibility has been carried out. The system is technically feasible for development and can be developed with the existing facility.

REQUEST APPROVAL

Not all request projects are desirable or feasible. Some organization receives so many project requests from client users that only few of them are pursued. However, those projects that are both feasible and desirable should be put into schedule. After a project request is approved, it cost, priority, completion time and personnel requirement is estimated and used to determine where to add it to any project list. Truly speaking, the approval of those above factors, development works can be launched.

CHAPTER 3
PROJECT DESIGN

3.1 INPUT DESIGN

Input Design plays a vital role in the life cycle of software development, it requires very careful attention of developers. The input design is to feed data to the application as accurate as possible. So inputs are supposed to be designed effectively so that the errors occurring while feeding are minimized. According to Software Engineering Concepts, the input forms or screens are designed to provide to have a validation control over the input limit, range and other related validations.

This system has input screens in almost all the modules. Error messages are developed to alert the user whenever he commits some mistakes and guides him in the right way so that invalid entries are not made. Let us see deeply about this under module design.

Input design is the process of converting the user created input into a computer-based format. The goal of the input design is to make the data entry logical and free from errors. The error is in the input are controlled by the input design. The application has been developed in user-friendly manner. The forms have been designed in such a way during the processing the cursor is placed in the position where must be entered. The user is also provided with in an option to select an appropriate input from various alternatives related to the field in certain cases.

Validations are required for each data entered. Whenever a user enters an erroneous data, error message is displayed and the user can move on to the subsequent pages after completing all the entries in the current page.

3.2 OUTPUT DESIGN

The Output from the computer is required to mainly create an efficient method of communication within the company primarily among the project leader and his team members, in other words, the administrator and the clients. The output of VPN is the system which allows the project leader to manage his clients in terms of creating new clients and assigning new projects to them, maintaining a record of the project validity and providing folder level access to each client on the user side depending on the projects allotted to him. After completion of a project, a new project may be assigned to the client. User authentication procedures are maintained at the initial stages itself. A new user may be created by the administrator himself or a user can himself register as a new user but the task of assigning projects and validating a new user rests with the administrator only.

The application starts running when it is executed for the first time. The server has to be started and then the internet explorer in used as the browser. The project will run on the local area network so the server machine will serve as the administrator while the other connected systems can act as the clients. The developed system is highly user friendly and can be easily understood by anyone using it even for the first time.

3.3 SYSTEM ARCHITECTURE

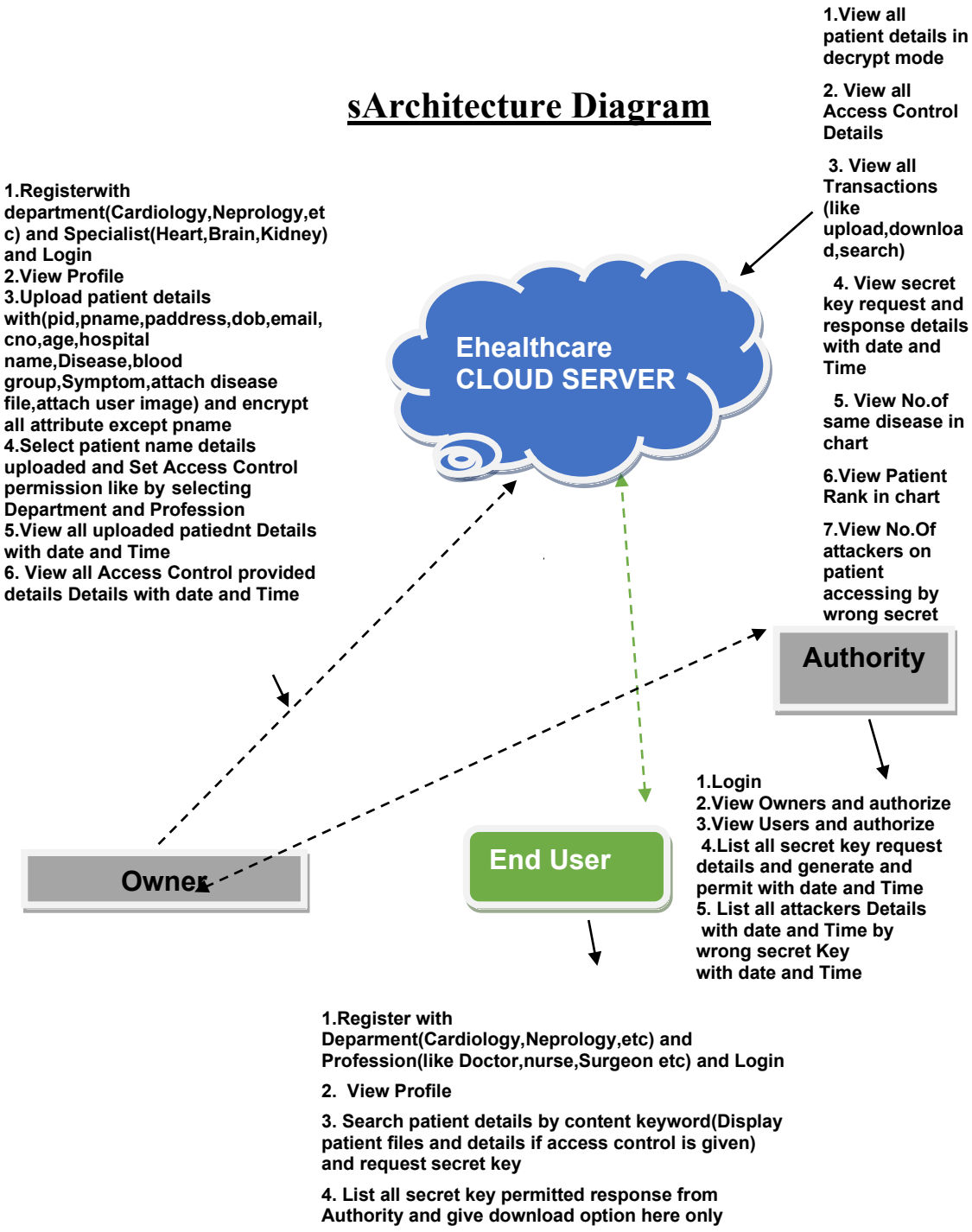


Fig:3.3:System Architecture

3.4 Data flow diagram

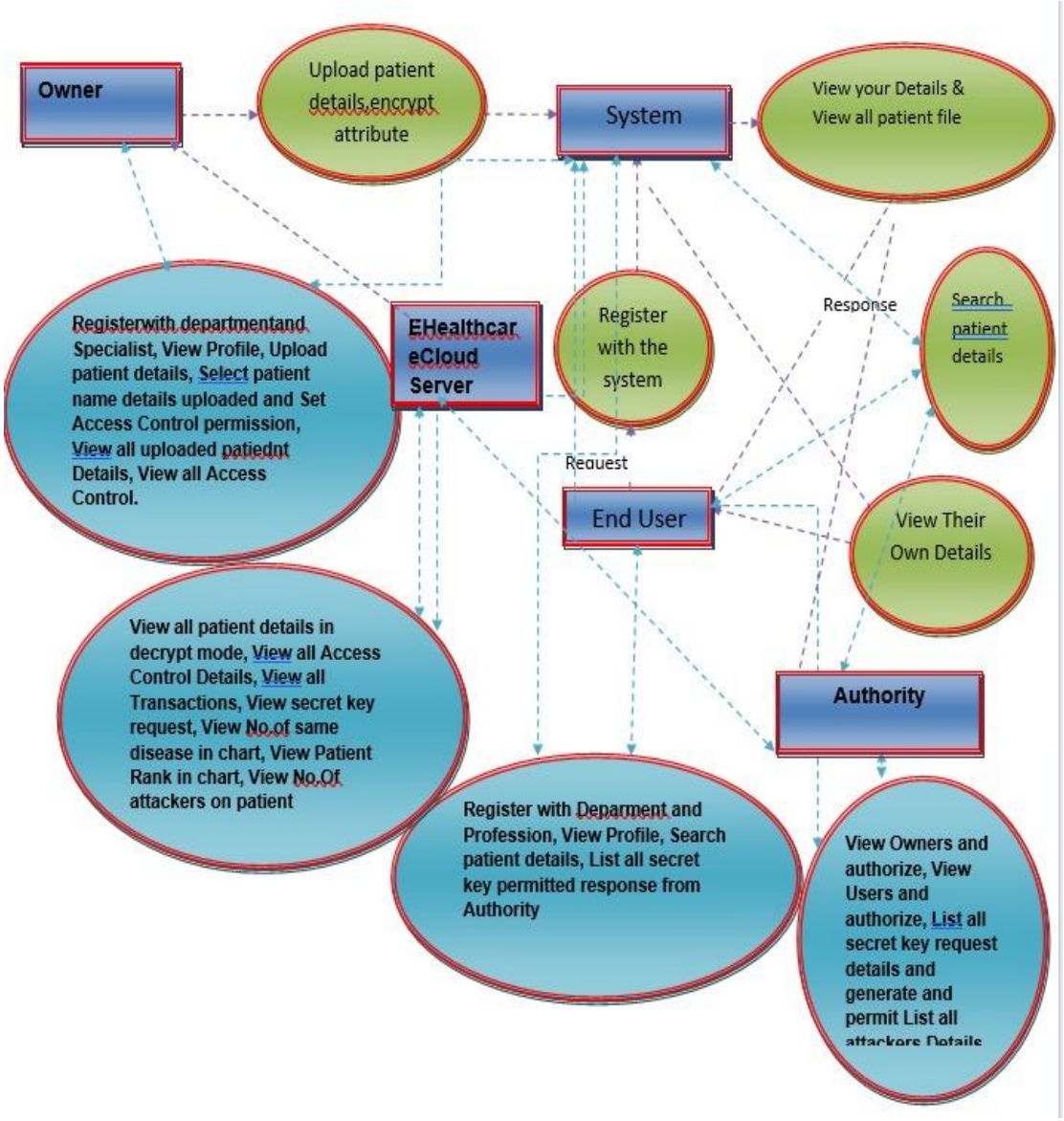


Fig:3.4 Data flow diagram

3.4.1 FLOW CHART

FLOW CHART : USER

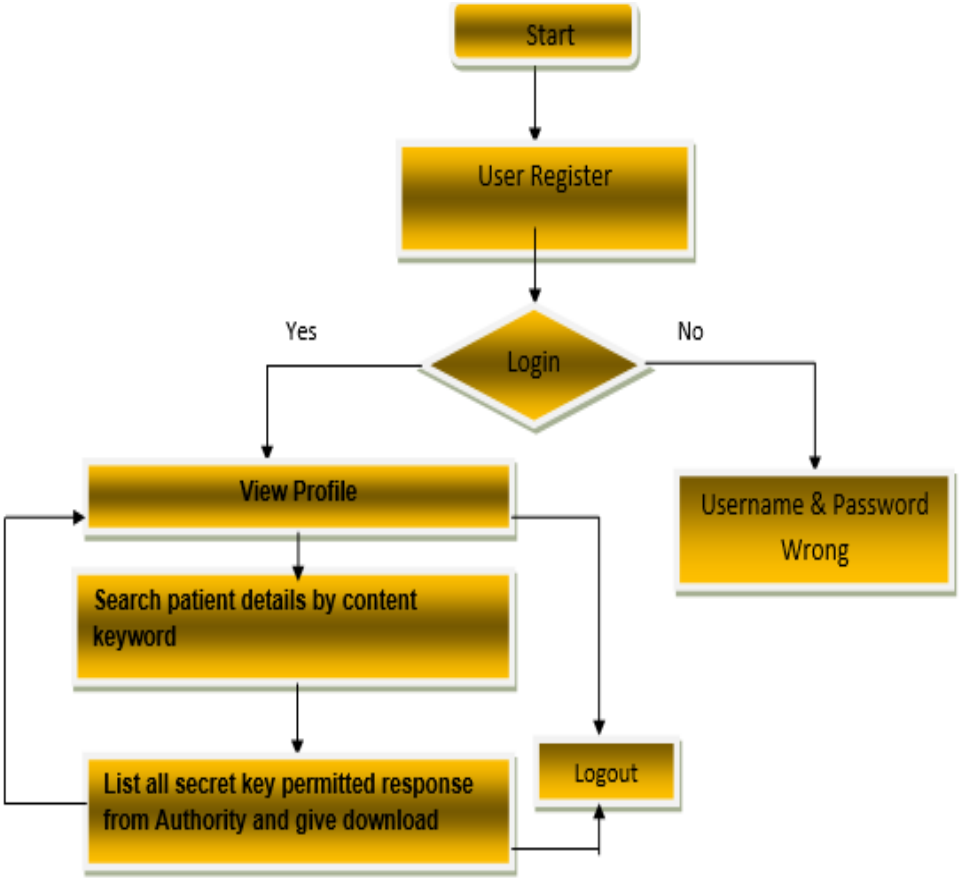


Fig:3.4.1.1:User Flow Chart

FLOW CHART : EHEALTH CLOUD SERVER

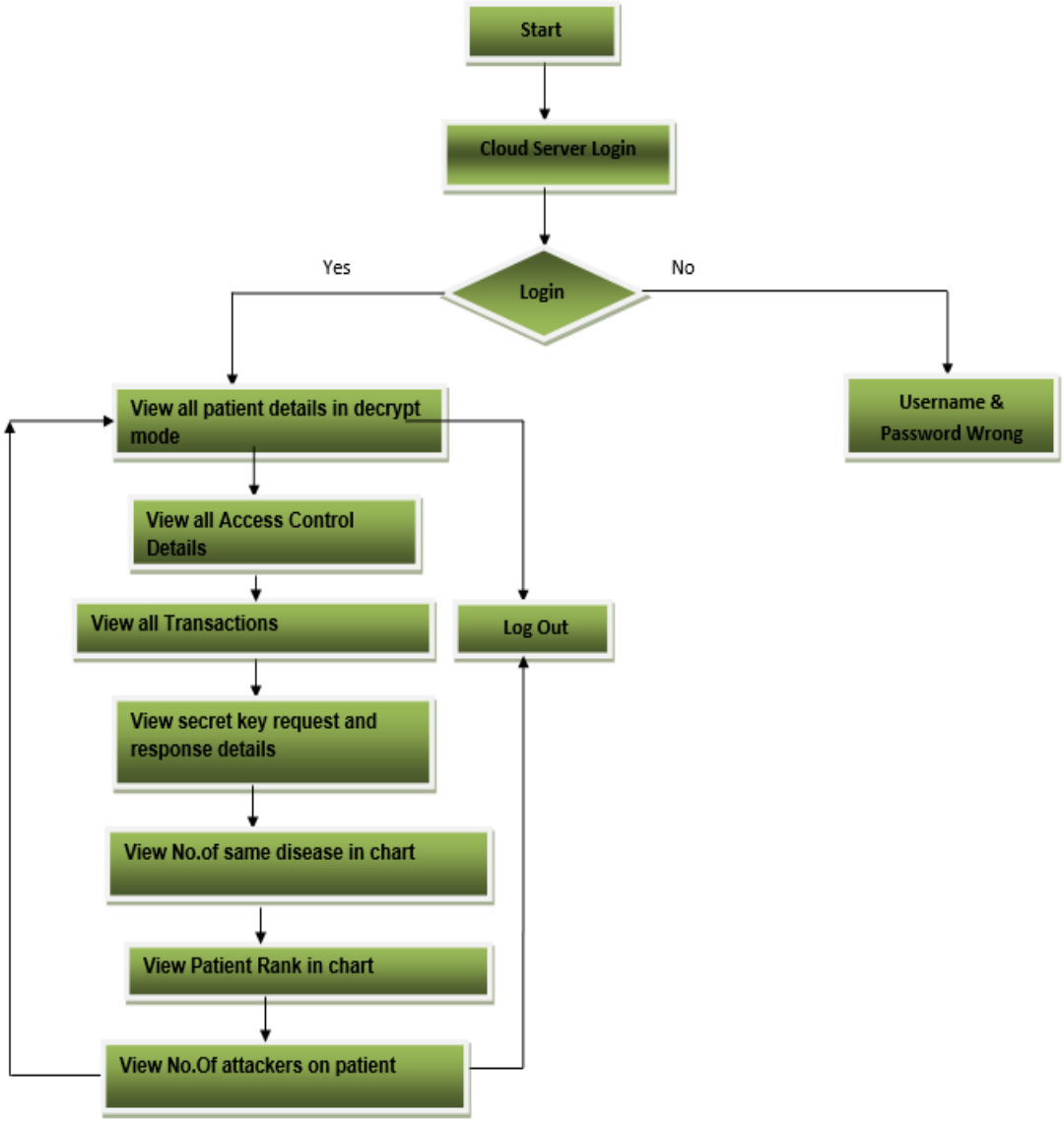


Fig:3.4.1.2 EHealth Cloud Server

FLOW CHART : AUTHORITY

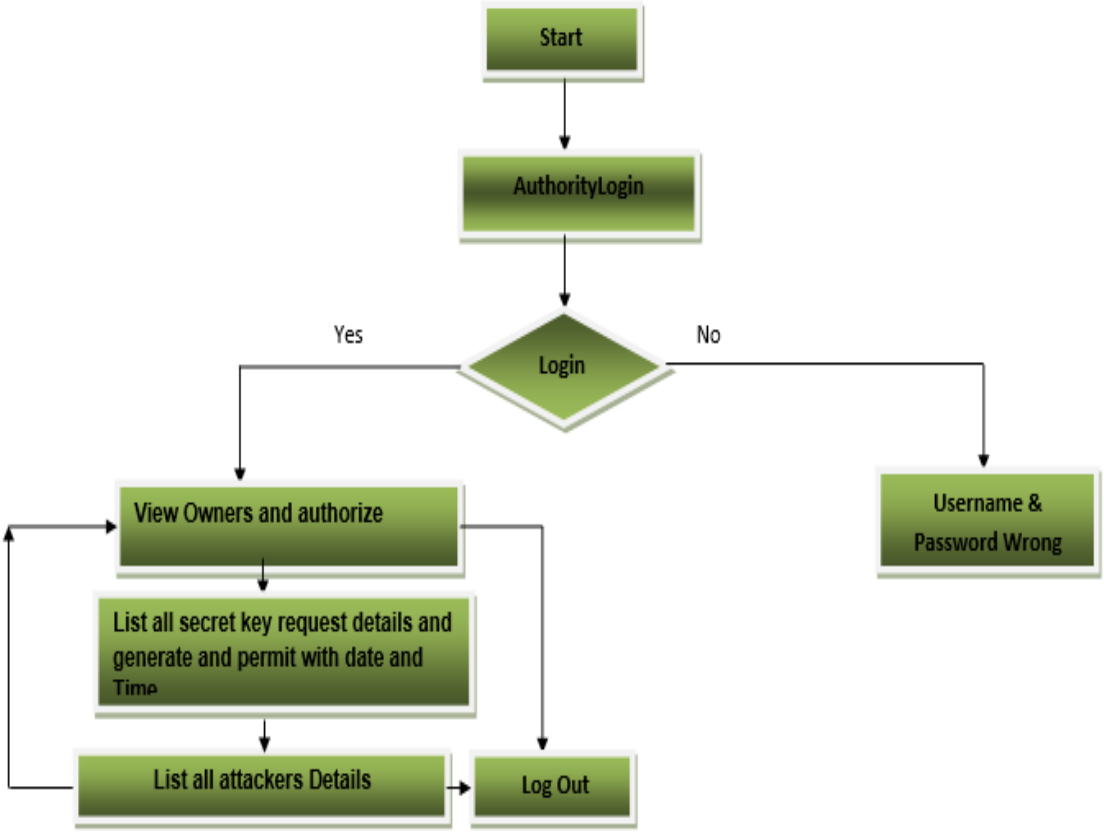


Fig:3.4.1.3 Authority Flowchart

FLOW CHART : OWNER

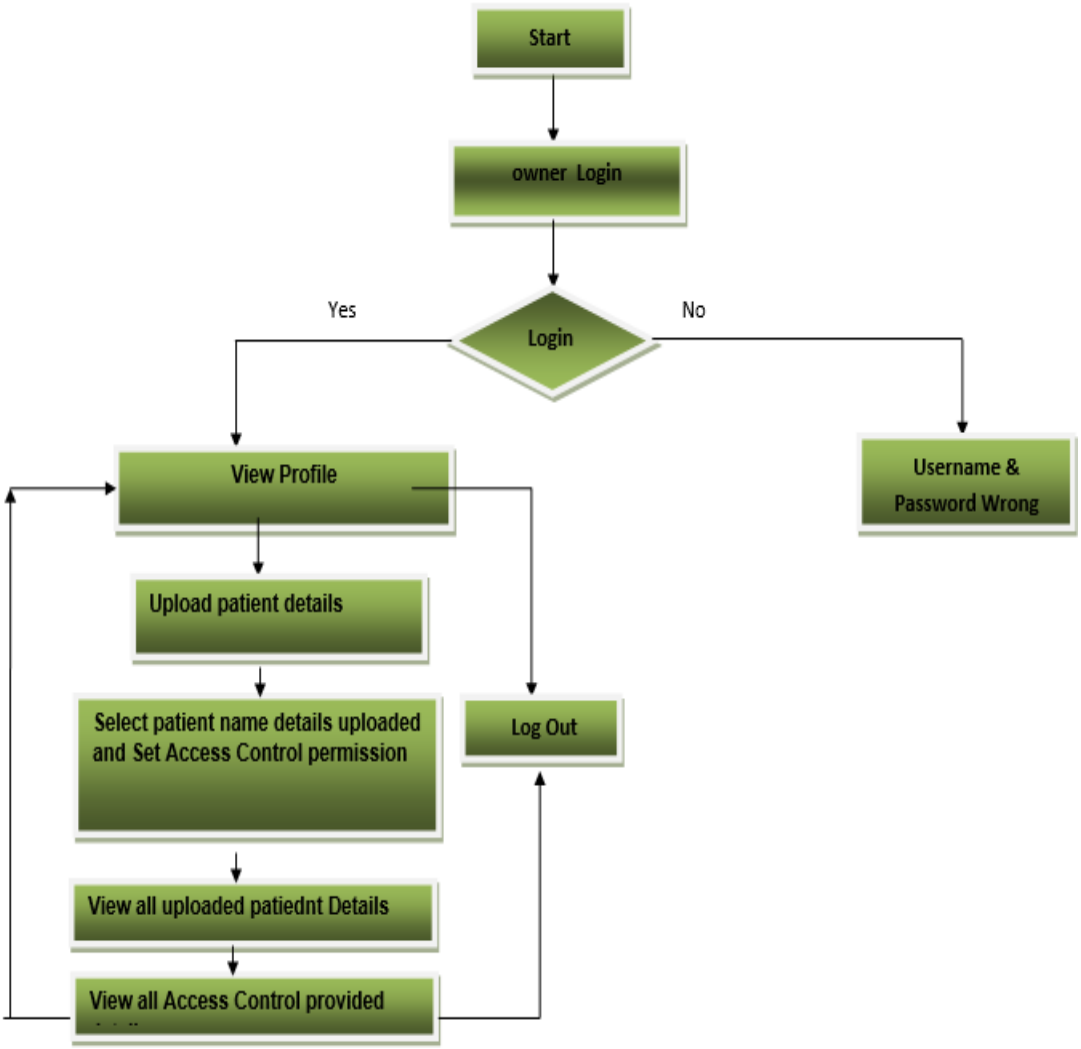


Fig:3.4.1.4 Owner Flowchart

3.5 UNIFIED MODELLING LANGUUAGE

UML stands for Unified Modeling Language. UML is a standardized general purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML. The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems. The UML is a very important part of developing objects-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

3.5 UML DIAGRAMS

A UML diagram is a diagram based on the UML (Unified Modeling Language) with the purpose of visually representing a system along with its main actors, roles, actions, artifacts or classes, in order to better understand, alter, maintain, or document information about the system

3.5.1 USE CASE DIAGRAM :

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

- **Use case** :Cloud Server

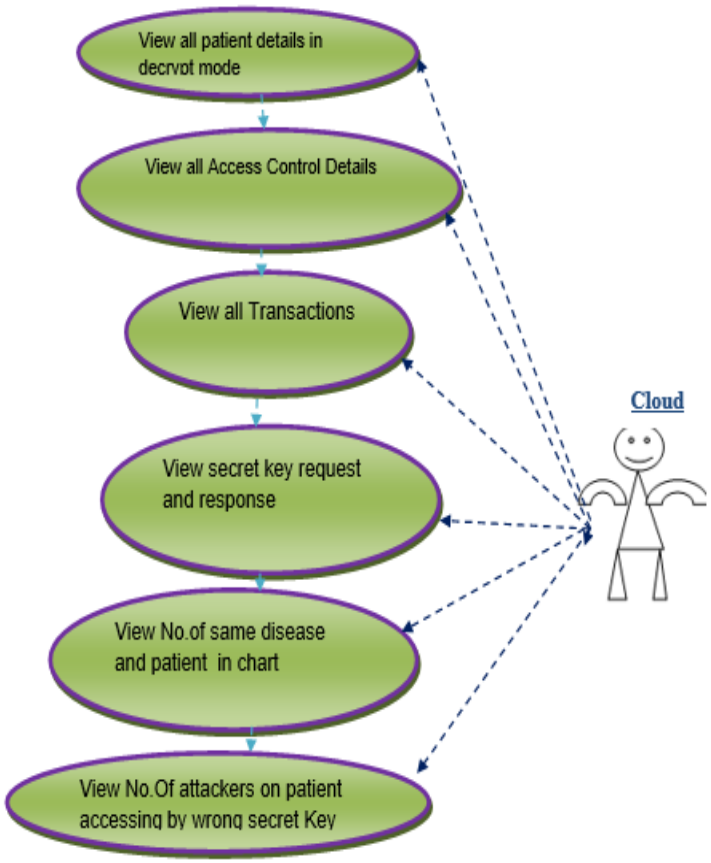


Fig:3.5.1.1 Cloud Server Use Case diagram

Use case : Authority

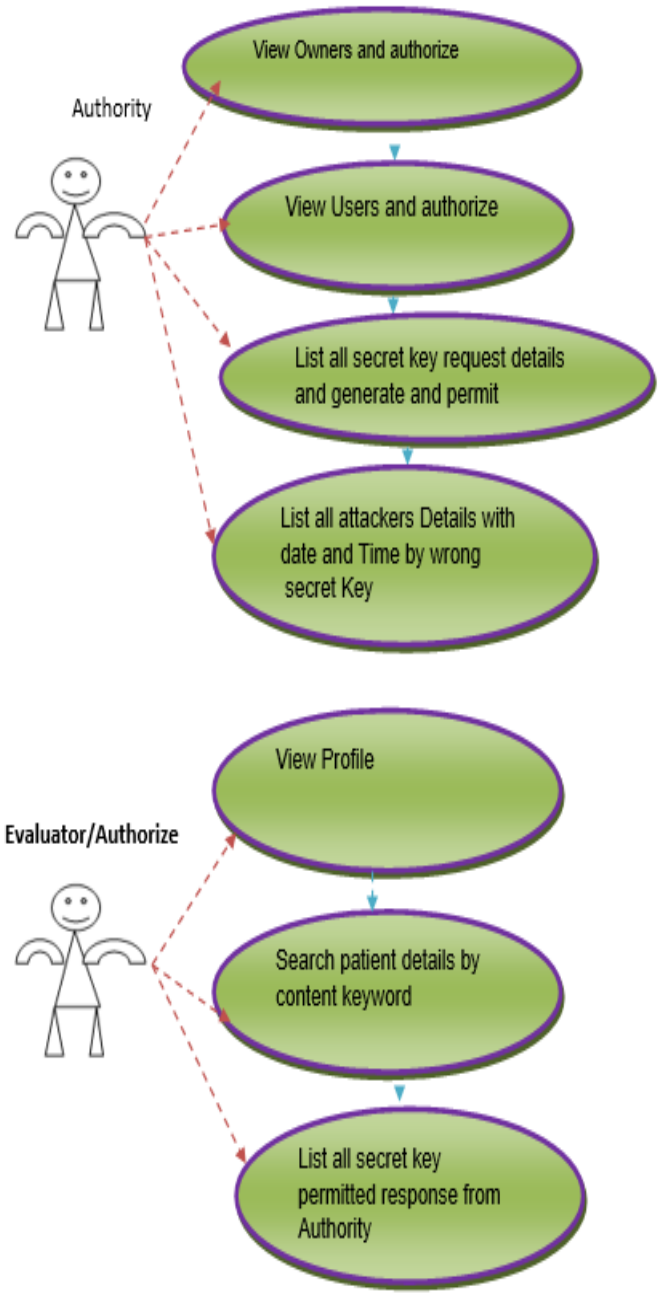


Fig:3.5.1.2 Authority Use Case diagram

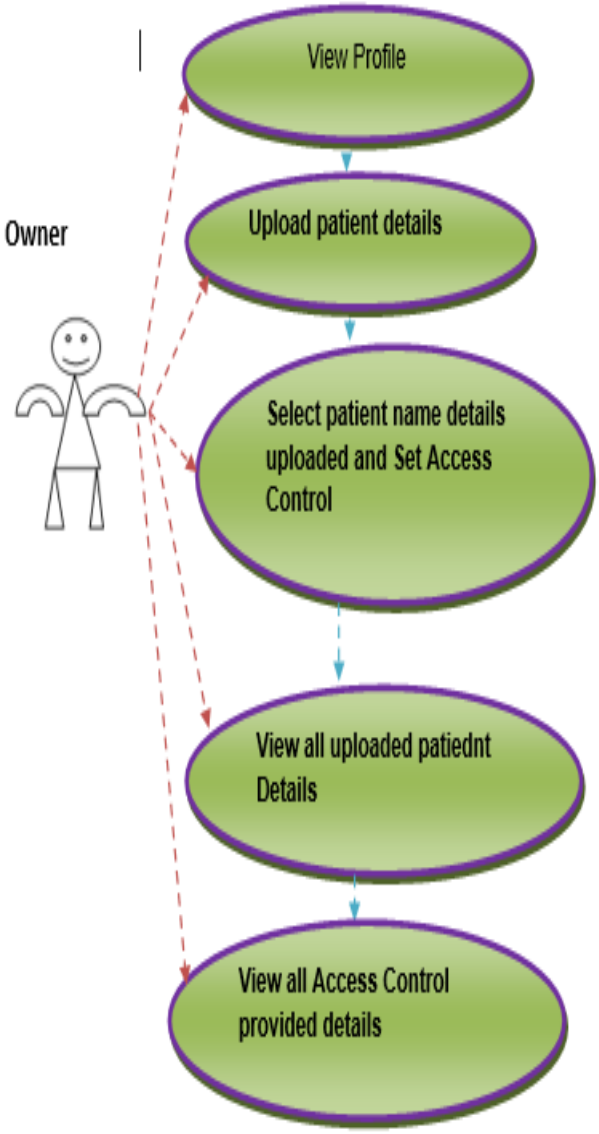
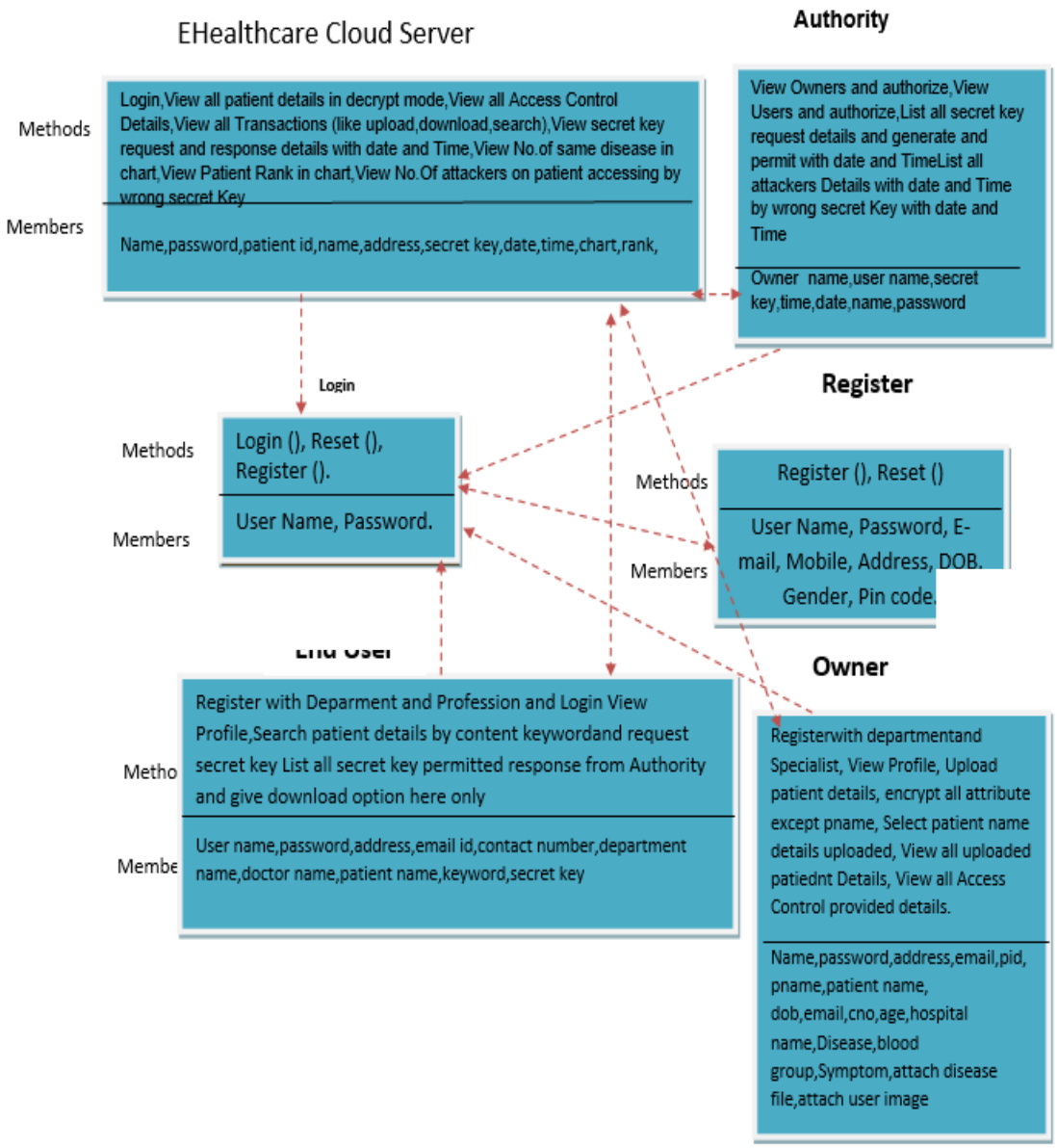


Fig:3.5.1.3 Owner Use Case diagram

3.5.2 CLASS DIAGRAM :



3.5.2 class diagram

3.5.3 SEQUENCE DIAGRAM :

- **Sequence Diagram**

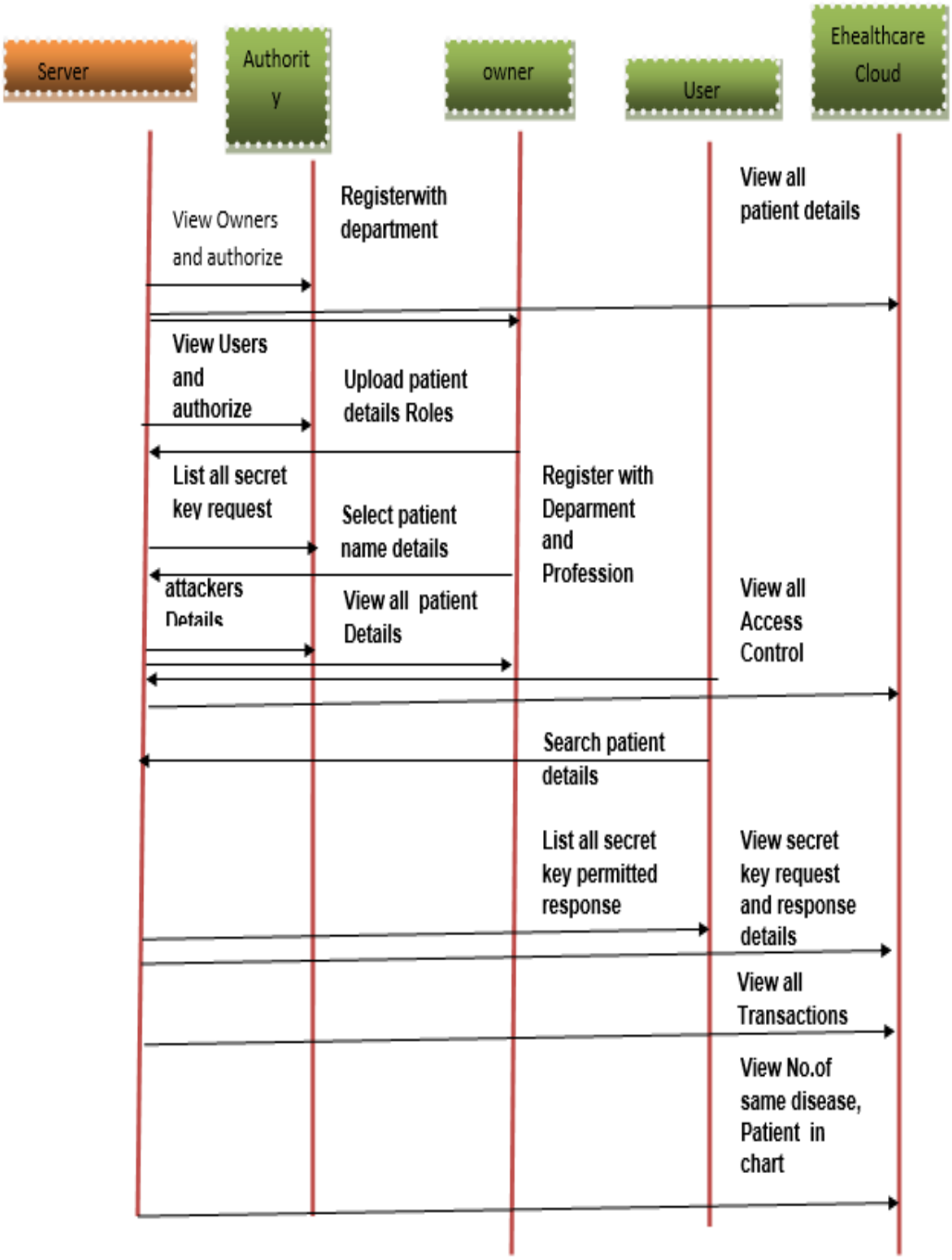


Fig:3.5.3 Sequence diagram

CHAPTER 4

IMPLEMENTATION

DATA OWNER

In this module, Data owner has to register to cloud and logs in, Encrypts and uploads a file to cloud server and also performs the department(Cardiology,Neprology,etc),Specialist(Heart,Brain,Kidney) with(pid,pname,paddress,dob,email,cno,age,hospital,name,Disease,blo od group,Symptom,attach disease file, attach user image) and encrypt all attribute except pname ,Select patient name details uploaded and Set Access Control permission like by selecting Department and Profession and View all uploaded patient Details with date and Time ,View all Access Control provided details with date and Time.

Ehealthcare CLOUD SERVER

In this module the cloud will authorize both the owner and the user and also performs the following operations such as View all patient details in decrypt mode and View all Access Control Details, View all Transactions (like upload, download, search) and View secret key request and response details with date and Time View No.of same disease in chart, View Patient Rank in chart and View No.Of attackers on patient accessing by wrong secret Key

Authority

Time and List all attackers Details with date and Time by wrong secret Key with date and Time. In this module, the Authority performs the following operations such as Login ,view Owners and authorize and View Users and authorize,List all secret key request details and generate and permit with date and

End USER

In this module, the user has to register to cloud and log in and performs the following operations such as Register with Deparment(Cardiology,Neprology,etc)andProfession(likeDoctor,nurse,Surgeon etc) and Login ,View Profile and Search patient details by content keyword(Display patient files and details if access control is given) and request secret key and List all secret key permitted response from Authority and give download option here only.

CHAPTER 5

RESULTS

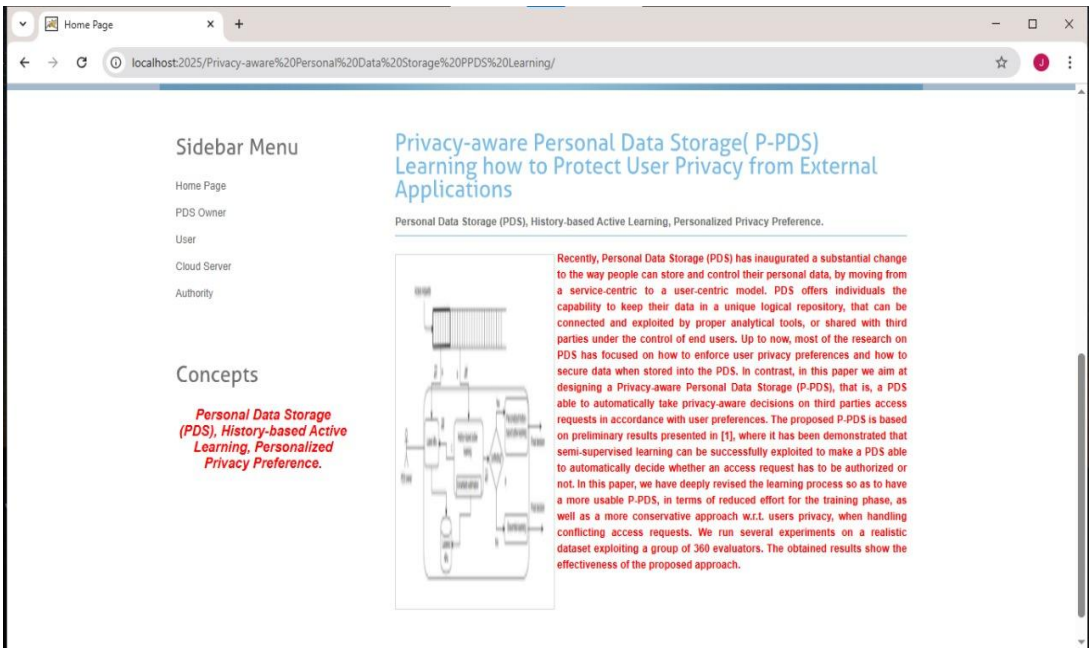


Fig:5.1 Home Page

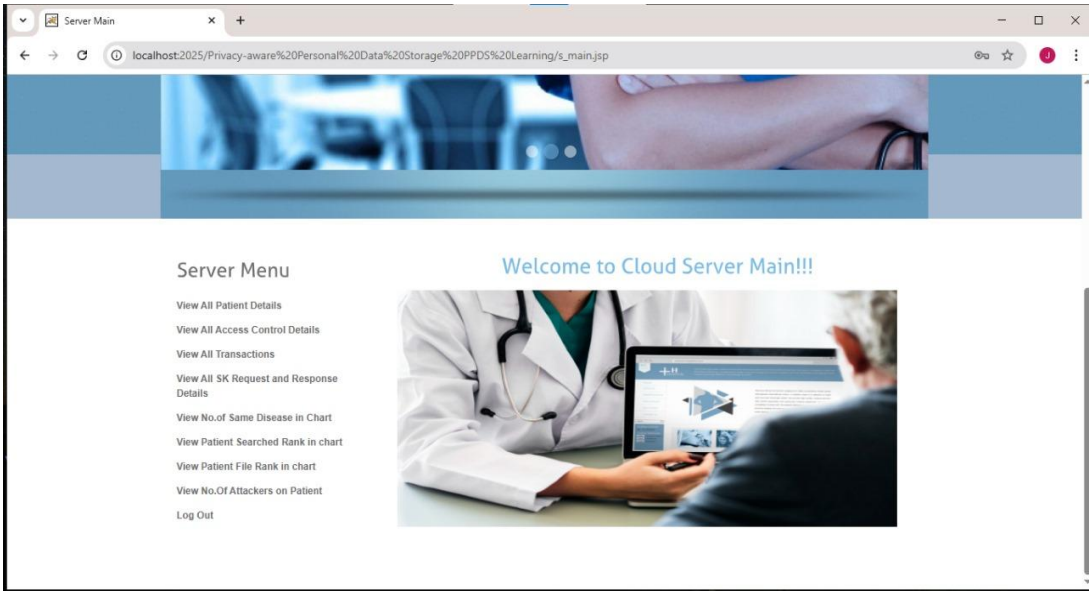


Fig: 5.2 cloud server menu

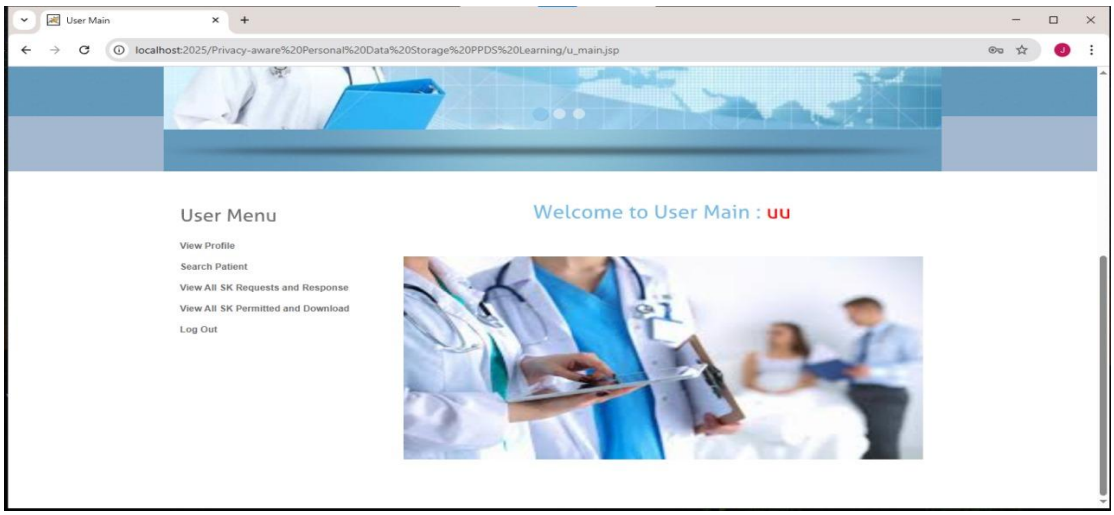


Fig:5.3 user menu

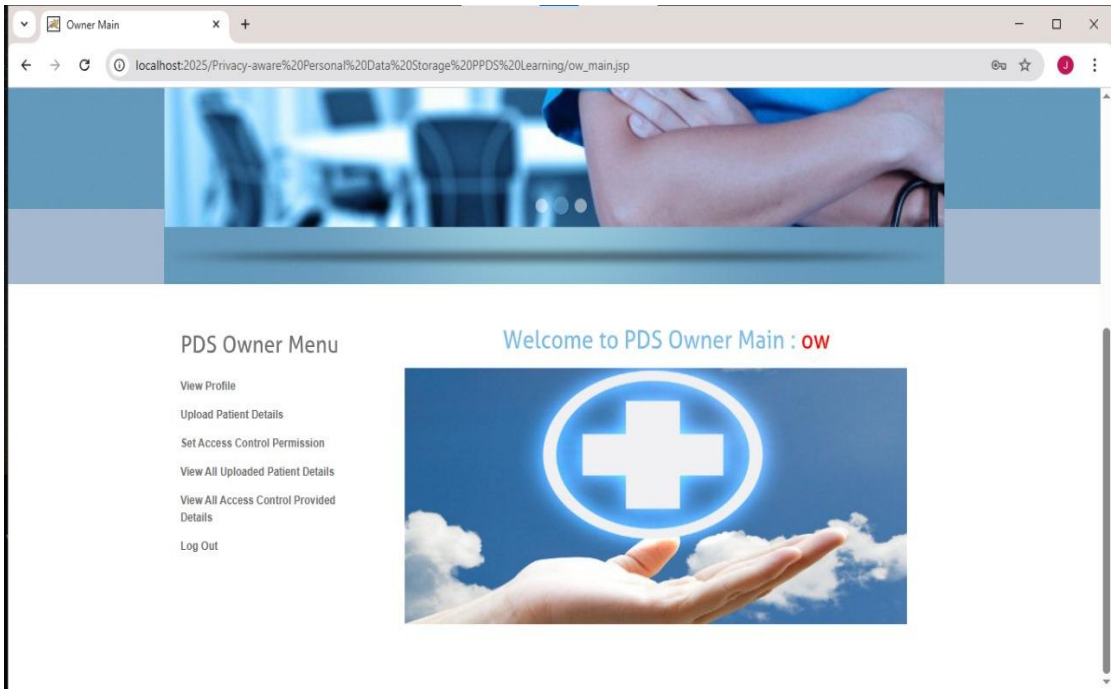


Fig: 5.4 PDS owner menu

Fig: 5.5 Authority menu

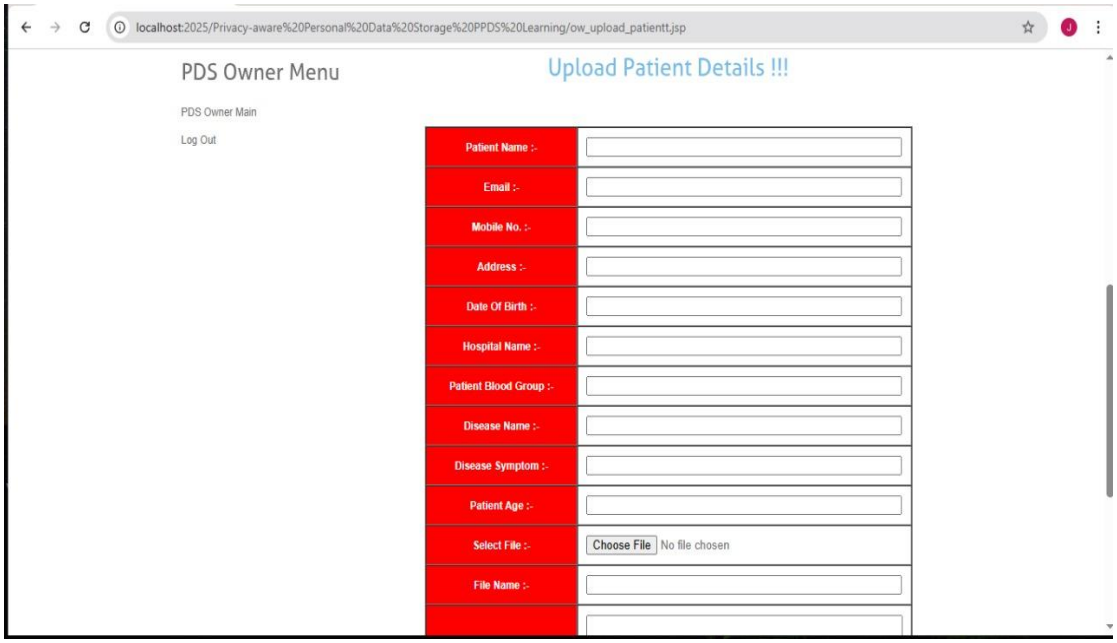


Fig :5.6 upload patient details

PDS Owner Menu

PDS Owner Main

Log Out

Upload Patient Details

Patient Name :-	anshu
Email :-	an@gmail.com
Mobile No. :-	9445634278
Address :-	wrgl
Date Of Birth :-	6/6/2000
Hospital Name :-	padhma
Patient Blood Group :-	b+
Disease Name :-	cancer
Disease Symptom :-	hair fall
Patient Age :-	25
File Name :-	a.txt
	PK[illegible][Content_Types].xml [illegible]

Fig: 5.7 patient details

Patient Uploaded Successfully !!!

BACK

Fig: 5.8 uploaded succesfully

CHAPTER 6
TESTING

6.1 SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the

Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

6.2 TYPES OF TESTS

6.2.1 Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

6.2.2 Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

6.2.3 Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.
Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures: interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

6.2.4 System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

6.2.5 White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

6.2.6 Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing:

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

SYSTEM TESTING

TESTING METHODOLOGIES

The following are the Testing Methodologies:

- **Unit Testing.**
- **Integration Testing.**
- **User Acceptance Testing.**
- **Output Testing.**
- **Validation Testing.**

Unit Testing

Unit testing focuses verification effort on the smallest unit of Software design that is the module. Unit testing exercises specific paths in a module's control structure to ensure complete coverage and maximum error detection. This test focuses on each module individually, ensuring that it functions properly as a unit. Hence, the naming is Unit Testing.

During this testing, each module is tested individually and the module interfaces are verified for the consistency with design specification. All important processing path are tested for the expected results. All error handling paths are also tested.

Integration Testing

Integration testing addresses the issues associated with the dual problems of verification and program construction. After the software has been integrated a set of high order tests are conducted. The main objective in this testing process is to take unit tested modules and builds a program structure that has been dictated by design.

The following are the types of Integration Testing:

1)Top Down Integration

This method is an incremental approach to the construction of program structure. Modules are integrated by moving downward through the control hierarchy, beginning with the main program module. The module subordinates to the main program module are incorporated into the structure in either a depth first or breadth first manner.

In this method, the software is tested from main module and individual stubs are replaced when the test proceeds downwards.

2. **Bottom-up Integration**

This method begins the construction and testing with the modules at the lowest level in the program structure. Since the modules are integrated from the bottom up, processing required for modules subordinate to a given level is always available and the need for stubs is eliminated. The bottom up integration strategy may be implemented with the following steps:

- The low-level modules are combined into clusters into clusters that perform a specific Software sub-function.
- A driver (i.e.) the control program for testing is written to coordinate test case input and output.
- The cluster is tested.
- Drivers are removed and clusters are combined moving upward in the program structure

The bottom up approaches tests each module individually and then each module is module is integrated with a main module and tested for functionality.

6.3 OTHER TESTING METHODOLOGIES

User Acceptance Testing

User Acceptance of a system is the key factor for the success of any system. The system under consideration is tested for user acceptance by constantly keeping in touch with the prospective system users at the time of developing and making changes wherever required. The system developed provides a friendly user interface that can easily be understood even by a person who is new to the system.

Output Testing

After performing the validation testing, the next step is output testing of the proposed system, since no system could be useful if it does not produce the required output in the specified format. Asking the users about the format required by them tests the outputs generated or displayed by the system under consideration. Hence the output format is considered in 2 ways – one is on screen and another in printed format.

Validation Checking

Validation checks are performed on the following fields.

Text Field:

The text field can contain only the number of characters lesser than or equal to its size. The text fields are alphanumeric in some tables and alphabetic in other tables. Incorrect entry always flashes and error message.

Numeric Field:

The numeric field can contain only numbers from 0 to 9. An entry of any character flashes an error messages. The individual modules are checked for accuracy and what it has to perform. Each module is subjected to test run along with sample data. The individually tested modules are integrated into a single system.

Testing involves executing the real data information is used in the program the existence of any program defect is inferred from the output. The testing should be planned so that all the requirements are individually tested.

A successful test is one that gives out the defects for the inappropriate data and produces and output revealing the errors in the system.

Preparation of Test Data

Taking various kinds of test data does the above testing. Preparation of test data plays a vital role in the system testing. After preparing the test data the system under study is tested using that test data. While testing the system by using test data errors are again uncovered and corrected by using above testing steps and corrections are also noted for future use.

Using Live Test Data:

Live test data are those that are actually extracted from organization files. After a system is partially constructed, programmers or analysts often ask users to key in a set of data from their normal activities. Then, the systems person uses this data as a way to partially test the system. In other instances, programmers or analysts extract a set of live data from the files and have them entered themselves.

It is difficult to obtain live data in sufficient amounts to conduct extensive testing. And, although it is realistic data that will show how the system will perform for the typical processing requirement, assuming that the live data entered are in fact typical, such data generally will not test all combinations or formats that can enter the system. This bias toward typical values then does not provide a true systems test and in fact ignores the cases most likely to cause system failure.

Using Artificial Test Data:

Artificial test data are created solely for test purposes, since they can be generated to test all combinations of formats and values. In other words, the artificial data, which can quickly be prepared by a data generating utility program in the information systems department, make possible the testing of all login and control paths through the program.

The most effective test programs use artificial test data generated by persons other than those who wrote the programs. Often, an independent team of testers formulates a testing plan, using the systems specifications.

The package "Virtual Private Network" has satisfied all the requirements specified as per software requirement specification and was accepted.

USER TRAINING

Whenever a new system is developed, user training is required to educate them about the working of the system so that it can be put to efficient use by those for whom the system has been primarily designed. For this purpose the normal working of the project was demonstrated to the prospective users. Its working is easily understandable and since the expected users are people who have good knowledge of computers, the use of this system is very easy.

MAINTAINENCE

This covers a wide range of activities including correcting code and design errors. To reduce the need for maintenance in the long run, we have more accurately defined the user's requirements during the process of system development. Depending on the requirements, this system has been developed to satisfy the needs to the largest possible extent. With development in technology, it may be possible to add many more features based on the requirements in future. The coding

and designing is simple and easy to understand which will make maintenance easier.

TESTING STRATEGY :

A strategy for system testing integrates system test cases and design techniques into a wellplanned series of steps that results in the successful construction of software. The testing strategy must co-operate test planning, test case design, test execution, and the resultant data collection and evaluation. A strategy for software testing must accommodate low-level tests that are necessary to verify that a small source code segment has been correctly implemented as well as high level tests that validate major system functions against user requirements.

Software testing is a critical element of software quality assurance and represents the ultimate review of specification design and coding. Testing represents an interesting anomaly for the software. Thus, a series of testing are performed for the proposed system before the system is ready for user acceptance testing.

SYSTEM TESTING:

Software once validated must be combined with other system elements (e.g. Hardware, people, database). System testing verifies that all the elements are proper and that overall system function performance is achieved. It also tests to find discrepancies between the system and its original objective, current specifications and system documentation.

UNIT TESTING:

In unit testing different are modules are tested against the specifications produced during the design for the modules. Unit testing is essential for verification of the code produced during the coding phase, and hence the goals to test the internal logic of the modules. Using the detailed design description as a guide, important

Conrail paths are tested to uncover errors within the boundary of the modules. This testing is carried out during the programming stage itself. In this type of testing step, each module was found to be working satisfactorily as regards to the expected output from the module.

In Due Course, latest technology advancements will be taken into consideration. As part of technical build-up many components of the networking system will be generic in nature so that future projects can either use or interact with this. The future holds a lot to offer to the development and refinement of this project.

CHAPTER 7
CONCLUSION

This paper proposes a Privacy-aware Personal Data Storage, able to automatically take privacy aware decisions on third parties access requests in accordance with user preferences. The system relies on active learning complemented with strategies to strengthen user privacy protection. As discussed in the paper, we run several experiments on a realistic dataset exploiting a group of 360 evaluators. The obtained results show the effectiveness of the proposed approach. We plan to extend this work along several directions. First, we are interested to investigate how P-PDS could scale in the IoT scenario, where access requests decision might depend also on contexts, not only on user preferences. Also, we would like to integrate P-PDS with cloud computing services (e.g., storage and computing) so as to design a more powerful P-PDS by, at the same time, protecting users privacy.

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